



SIWESync AI

Team Documentation & Technical Reference

Syncing Students with Opportunity

By Team Avengers

Version 1.0 · Hackathon Build

This document is the complete reference for every teammate working on SIWESync AI. It covers the project vision, all six AI features, the full tech stack, codebase structure, the 30-company internship database, API integration, deployment guide, and the hackathon pitch strategy. Read this before touching the code.

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01 Project Overview & Problem Statement

SIWESync AI is an AI-powered platform that matches Nigerian university students with SIWES (Students Industrial Work Experience Scheme) internship placements, evaluates their readiness, identifies skill gaps, and generates a personalised action plan to make them competitive. Built as a fully functional hackathon demo.

The Problem

- Students apply for SIWES placements blindly — no guidance on which companies suit their profile.
- Rejection is common because students don't know what skills companies actually require.
- Students receive no feedback, leaving them stuck in a cycle of unsuccessful applications.
- Universities provide no structured preparation pathway before the placement period.
- Companies struggle to find well-prepared interns and receive unqualified applicants.

The Opportunity

- Thousands of Nigerian students undergo SIWES every year — it is a mandatory university requirement.
- The gap between student readiness and industry expectation is large and measurable.
- AI can personalise guidance at scale in a way no career office can.
- A platform solving this has a clear path to a full recruitment and upskilling product.



02 Solution & Core Value Proposition

SIWESync AI takes a student's CV and target role, runs it through Claude AI, and produces a complete career readiness report in under 30 seconds — covering six dimensions.

User Flow (End to End)

1	Student opens the SIWESync AI website
2	Optionally enters target role (e.g. 'Software Engineering') and weeks until SIWES starts
3	Uploads CV as PDF / .txt / .md — or pastes text directly into the textarea
4	Clicks 'Analyse My Profile' — progress bar shows real-time steps

5	Claude AI processes CV against 30 Nigerian company profiles
6	Dashboard loads with 6 tabs: Matches · Readiness · Skill Gaps · Roadmap · CV Upgrade · Interview Prep
7	Student uses the report to apply to matched companies and follow the roadmap

Why This Wins at the Hackathon

- **Real problem:** Every judge knows a student who struggled with SIWES placement.
- **Live demo:** Upload a real CV and show the full dashboard in under 30 seconds.
- **Technical depth:** Claude AI reasoning, PDF parsing, multi-tab dashboard, real company data.
- **Business case:** Clear path to B2B (universities, companies) and B2C (students) revenue.
- **Nigerian context:** Built specifically for the Nigerian educational system — not generic.

03 Tech Stack

Layer	Technology	Purpose
Frontend	Next.js 14 (App Router)	React framework — routing, SSR, API routes
Styling	Tailwind CSS + inline styles	Dark theme UI with custom CSS variables
Fonts	Syne + DM Sans (Google Fonts)	Display font + body font loaded via <link>
Backend	Next.js API Routes	Serverless — no separate server needed
AI Engine	Claude claude-opus-4-5 via Anthropic SDK	Matching, scoring, roadmap, CV, interview generation
PDF Parsing	pdf-parse v1.1.1	Extracts text from uploaded PDF CVs
Internship Data	JSON file (data/internships.json)	30 Nigerian company profiles
Icons	lucide-react	UI icons throughout the dashboard
Package Manager	npm	Dependency management
Deployment	Vercel (recommended)	One-click deploy, free tier, env vars support

Key Design Decisions

- **No separate backend server:** Next.js API routes handle everything — simpler to run and deploy.
- **Mock JSON dataset:** Reliable for demos; no external API dependencies that could fail during the pitch.
- **pdf-parse imported via lib path:** Avoids a v1 bug where the main entry crashes by trying to open a test file that doesn't exist in installations.
- **All in one Next.js app:** Teammates can clone, install, and run with two commands.

04 Codebase Structure

Standard Next.js 14 application using the App Router. Every file a teammate needs to understand is described below.



siwesync/	Root project folder
■■■ app/	Next.js App Router directory
■ ■■■■ page.tsx	Landing page — hero, upload form, loading state
■ ■■■■ layout.tsx	Root layout — fonts, favicon, metadata
■ ■■■■ globals.css	Design system — CSS variables, animations
■■■■■ api/analyze/	API route directory
■ ■■■■ route.ts	POST /api/analyze — PDF parsing → Claude → JSON
■■■ components/	Shared React components
■ ■■■■ Dashboard.tsx	6-tab results dashboard — all feature UIs
■■■ data/	Static data
■ ■■■■ internships.json	30 Nigerian company profiles for matching
■■■ public/	Static assets served at /
■ ■■■■ logo.png	SIWESync AI logo — nav, dashboard, browser tab
■■■ .env.local	ANTHROPIC_API_KEY lives here — never commit
■■■ package.json	Dependencies and scripts
■■■ tsconfig.json	TypeScript configuration

Data Flow

Browser → page.tsx → POST /api/analyze → pdf-parse (if PDF) → Claude API → JSON response
→ Dashboard.tsx → 6 tabs rendered

05 All 6 AI Features — Detailed

Every feature is generated by a single Claude API call returning a structured JSON object. Dashboard.tsx reads the JSON and renders each tab.

01 Smart Internship Matching

Claude compares the student's skills and experience against all 30 companies. Returns top 3 matches with a match score (0–100), one-sentence reason, and full company details (location, stipend, duration, required skills). Displayed as cards with a match percentage bar and company colour coding.

Uses: CV text, internships dataset, target role

02 Readiness Score

A 0–100 score split into four sub-dimensions: Technical Skills, Experience, Education, and Project Work. An animated SVG ring draws itself on page load, colour-coded green through red. Labels: Beginner / Developing / Competitive / Strong / Exceptional. Animated bar charts show the breakdown.

Uses: CV text, skills extracted by Claude

03 Skill Gap Analysis

Claude identifies skills the student is missing relative to their target roles and matched companies. Each gap is tagged High / Medium / Low priority with a sentence explaining why it matters in context. Cards use colour-coded priority badges (red / amber / purple).

Uses: CV skills vs company requirements

04 AI Roadmap Generator

A 4-phase weekly plan (Week 1–2, 3–4, 5–6, 7–8) personalised to the student's gaps and weeks until SIWES. Each phase has a focus area, 2 actions, and 1–2 specific learning resources. Displayed as a vertical timeline with numbered nodes. Duration adjusts based on weeks input.

Uses: Skill gaps, weeks until SIWES, target role

05 CV Optimizer

Claude reviews each CV section and returns specific issues and suggested fixes — e.g. 'Projects section lacks measurable impact — add user counts or performance metrics.' Every suggestion is actionable, not vague. Cards show issue in red and the fix below.

Uses: Full CV text, section-by-section analysis

06 Interview Prep

Four questions (alternating Technical / Behavioral) tailored to the student's profile and matched companies. Each has a 'TIP' box explaining how to frame the answer. Colour coded: Technical = purple, Behavioral = teal.

Uses: Profile, matched companies, target role

Important: Single API Call

All 6 features are generated in one Claude API call. The prompt asks Claude to return a single structured JSON object containing all sections. This keeps latency low (one round-trip) and makes the analysis coherent — the roadmap references the same gaps the skill gap tab shows.

The JSON is parsed in route.ts, topMatches are enriched with full company data from internships.json, and the complete object is returned to the frontend in one response.

06 Claude AI Integration

Model Used

claude-opus-4-5 (claude-opus-4-5)

Opus 4 is chosen for its superior reasoning — it produces higher-quality skill gap analysis and roadmap suggestions compared to Haiku or Sonnet.

API Call Parameters

Parameter	Value	Why
model	claude-opus-4-5	Best reasoning quality for analysis tasks
max_tokens	4000	Enough for full JSON with all 6 sections
messages	Single user turn	Stateless — full context sent in one prompt
temperature	Default (1.0)	Not overridden; natural variation fine for this use case

Prompt Architecture

The prompt sent to Claude contains these sections in order:

Role	You are SIWESync AI, an expert Nigerian internship career advisor.
CV Text	The full extracted text from the student's CV.
Target Role	The optional role entered by the student (e.g. 'Software Engineering').
Time Context	How many weeks until SIWES — calibrates roadmap urgency and duration.
Internship Data	The full JSON of all 30 companies in the database.
Output Spec	Exact JSON schema Claude must return — no markdown, no preamble, no extra text.

JSON Response Schema (Abbreviated)

```
{ "studentName": "string", "studentSummary": "string", "readinessScore": 0-100,
  "readinessLabel": "Beginner | Developing | Competitive | Strong | Exceptional",
  "readinessBreakdown": { technicalSkills, experience, education, projectWork },
  "topMatches": [ { internshipId, company, role, matchScore, matchReason } ], "skillGaps": [
  { skill, importance: High|Medium|Low, context } ], "strengths": [ "string", "string",
  "string" ], "roadmap": [ { week, focus, actions[], resources[] } x4 phases ],
  "cvImprovements": [ { section, issue, suggestion } ], "interviewQuestions": [ { type:
  Technical|Behavioral, question, tip } x4 ], "motivationalMessage": "string" }
```



API Key Setup

Add your key to a `.env.local` file in the project root.

Format exactly: `ANTHROPIC_API_KEY=sk-ant-api03-xxxxxxx`

Never commit this file to git — it is already in `.gitignore`.

On Vercel: add it in Project Settings → Environment Variables.

07 Internship Database — 30 Companies

Lives at `data/internships.json`. Claude reads the full dataset during analysis and selects the top 3 matches based on the student's skills, target role, and experience level.

JSON Field Reference

Field	Type	Example
id	string	"001"
company	string	"Flutterwave"
role	string	"Software Engineering Intern"
location	string	"Lagos, Nigeria"
duration	string	"6 months"
sector	string	"Fintech"
skills_required	string[]	["JavaScript", "React", "Node.js"]
description	string	"Join Africa's leading payments company..."
stipend	string	"■80,000/month"
logo_initial	string	"F" (first letter shown on the match card)
color	string	"#FF5B1F" (brand colour used for the card accent)

Companies by Sector

Sector	Companies
Fintech	Flutterwave, Paystack, Interswitch, Cowrywise, SystemSpecs (Remita)
Tech	Andela, Google Nigeria (STEP)
Oil & Gas	NLNG, NNPC, TotalEnergies Nigeria, TotalEnergies EP Nigeria
Telecoms	MTN Nigeria, Airtel Nigeria, NCC (Regulator)
Banking	Access Bank, Zenith Bank, Stanbic IBTC
Engineering / Power	Siemens Nigeria, IKEDC / Eko DisCo, Julius Berger, ARM Harith
Manufacturing	Dangote Group, Lafarge Africa, Nigerian Breweries, Nestlé Nigeria
Healthcare / Science	LUTH Biomedical, NAFDAC, NiMet, NIMASA
E-Commerce	Konga

To add more companies: open `data/internships.json`, copy an existing entry, increment the id, and fill in the fields. The AI automatically considers new entries — no code changes needed.

08 API Reference



POST /api/analyze

The only API endpoint in the application.

Request — JSON body (pasted text)

```
{ "cvText": "string (min 50 chars)", "targetRole": "string (optional)", "weeksUntilSIWES":  
  "string (optional)" }
```

Request — multipart/form-data (file upload)

```
file: File (PDF / .txt / .md) | targetRole: string | weeksUntilSIWES: string
```

Success Response — 200 OK

Returns the full analysis JSON object (see schema in Section 06).

Error Responses

Status	Condition	Message returned
400	CV text too short or empty	"CV content is too short or could not be extracted..."
400	No file in form upload	"No file received."
500	API key missing or placeholder	"API key not configured. Add ANTHROPIC_API_KEY..."
500	Claude API error	"Analysis failed: <error message>"
500	JSON parse failure	"Analysis failed: <parse error>"

Critical: pdf-parse Import Path

Do Not Change This Import

pdf-parse v1 has a known bug: `require('pdf-parse')` triggers it to load a test PDF from its own `test/` folder on startup. That folder doesn't exist in installations, causing an `ENOENT` crash before your CV is even touched.

The fix in `route.ts`: `require('pdf-parse/lib/pdf-parse.js')` — skips the test loader entirely.

Do not change this import path. Do not upgrade to pdf-parse v2 (DOMMatrix browser-API issues).

09 Running Locally & Deployment

Prerequisites

- Node.js 18 or higher
- npm (bundled with Node.js)
- Anthropic API key — get one at console.anthropic.com

Local Setup

```
1. Unzip      unzip siwesync-demo.zip && cd siwesync
```

```
2. Set API key  echo "ANTHROPIC_API_KEY=sk-ant-..." > .env.local
```



3. Install

`npm install`
4. Run

`npm run dev`
5. Open

`http://localhost:3000`

Deploying to Vercel (Recommended)

- Push the project to a GitHub repository (exclude node_modules and .next).
- Go to vercel.com → New Project → Import your GitHub repo.
- In Project Settings → Environment Variables, add: ANTHROPIC_API_KEY = your key.
- Click Deploy. Vercel builds and deploys. You get a public URL instantly.
- Share the URL with judges — the demo runs live from any browser, no setup needed.

npm Scripts

Command	What it does
npm run dev	Start local development server with hot reload
npm run build	Build for production — also checks for TypeScript errors
npm run start	Serve the production build locally

10 Demo Strategy & Pitch Guide

The pitch is a live demo. Judges need to see AI working in real time. Follow this script for maximum impact.

Pitch Script (2–3 minutes)

OPENING	30s	"Every year, thousands of Nigerian students must complete SIWES. Most apply blindly, get rejected, and never know why. Companies need prepared interns but can't find them. That gap costs everyone."
SOLUTION	20s	"SIWESync AI is the bridge. Upload your CV — in under 30 seconds our AI matches you to the right company, scores your readiness, shows your exact skill gaps, and gives you a personalised 8-week plan to fix them."
DEMO	60s	Open the app live. Click "Use Sample CV". Set target role to "Software Engineering". Set weeks to 8w. Click Analyse. Walk through each tab as it loads: Matches → Readiness → Gaps → Roadmap → CV Upgrade → Interview Prep.
TECH	20s	"Next.js frontend, Claude claude-opus-4-5 for reasoning, pdf-parse for CV extraction, 30 real Nigerian companies in the database. One AI call — six outputs — under 30 seconds."



1 5 s	"This doesn't just help students apply — it helps them become qualified. Scale to universities and companies and you have a full recruitment platform."
IMP ACT	
1 0 s	"SIWESync AI — syncing students with opportunity."
CLO SE	

Demo Tips

- **Use the sample CV button** — guaranteed strong results, no risk of a weak real CV during the pitch.
- **Pre-load the page** before going on stage — first load fetches Google Fonts (~1 second delay).
- **Set weeks to 8w** — produces a clean, balanced roadmap for the demo.
- **Have a screenshot fallback** of the dashboard in case of internet issues.
- **Walk all 6 tabs** — each one is a new wow moment for the judges.

SIWESync AI — Built for the Hackathon — Syncing Students with Opportunity